

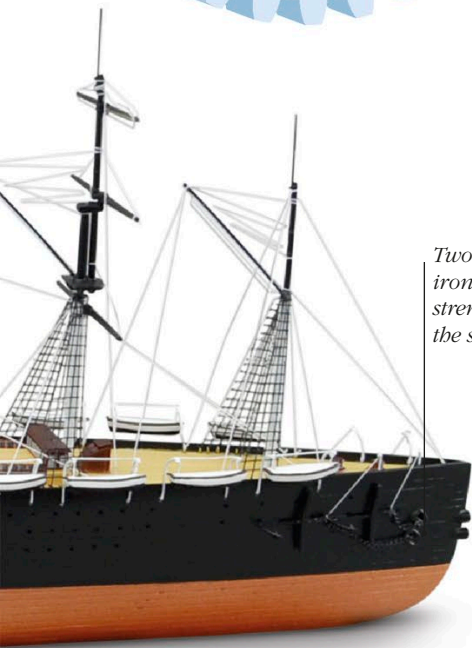
First turbine-powered boat

- **What?** *Turbinia*
- **Who?** Charles Parsons
- **Where and when?** UK, 1894

In 1884, the British engineer Charles Parsons invented the steam turbine, using high pressure steam to turn a turbine or fan. A decade later, he launched the first turbine-powered ship, the *Turbinia*. It was the fastest ship in the world at the time, reaching speeds of up to 40 mph (64 km/h). Most big ships today use diesel or diesel-electric engines.

WOW!

At 1,194 ft (364 m) long, the French-built *Harmony of the Seas* is the largest ever passenger ship.



Two-layered iron hull strengthened the ship.

Container ship

- **What?** *Ideal X*
- **Who?** Malcom McLean
- **Where and when?** US, 1956

In 1956, the American transportation tycoon Malcom McLean came up with the idea of putting loose cargo in

standard metal boxes or containers. These could be stacked high in specialized ships that would be easy to load and unload. The idea worked so well that most standard cargo is now carried in container ships.



MSC Agata, a container ship

Identically sized containers are stacked up on the deck of the ship.

GET MOVING

Largest solar-powered boat



- **What?** *Tûranor PlanetSolar*
- **Who?** Knierim Yachtbau
- **Where and when?** Germany, 2010

The *Tûranor PlanetSolar* is the world's largest solar-powered boat, and the first to sail around the world. It is driven by motors entirely powered by solar cells. The word "Tûranor" was invented by the British writer J. R. R. Tolkien in his famous book *The Lord of the Rings* and means "the power of the sun."

Robotic ship

- **What?** *YARA Birkeland*
- **Who?** Yara International
- **Where and when?** Norway, 2019

The *YARA Birkeland* will be the world's first autonomous ship—a robotic ship that can sail without a crew. It is also a zero-emission container ship that will carry fertilizers between small ports

in Norway. While it will have a small crew for the first year, it is expected to be able to run without any crew at all by 2020.

